

- CHAPTER 15 -

HEALTH AND INJURY MANAGEMENT

Injuries are a particularly hard topic to address since there is no one-size-fits-all prescription to nurse someone back to health. Everyone responds differently to treatments and has different recovery factors to take into account, including but not limited to: nutrition, sleep schedule, general overall health, and training schedule. It is absolutely crucial to receive individual care for your specific injury. Certain issues absolutely require a visit to an orthopedic sports doctor or physical therapist.

With this in mind, understand that the information contained in chapters fifteen and sixteen reflects a generalized approach to health and injury management. Something that may work very well for one athlete may not work at all for someone else—it could even have a negative effect! Every athlete is different. A doctor or therapist will be able to examine your individual situation and prescribe rehabilitative or prehabilitative work to restore you to health. If your medical professional says something that contradicts information provided in these chapters, listen to them as they know your case!

Any information found in chapters fifteen and sixteen of *Overcoming Gravity* is for informational purposes only. None of this should be misconstrued as medical advice. You should always consult your medical professional before utilizing any information located herein. Self-treatment without advice from a medical professional is not recommended and is therefore at your own risk.

ADDRESSING “THE BURN,” PAIN, AND SORENESS

Everyone has heard the saying, No pain, no gain. The phrase is ubiquitous in the athletic world for pushing through pain. Some coaches—typically at the high school level and below—believe in this saying unconditionally. Let’s distinguish between “the burn,” pain, and soreness in order to get a better understanding of the concepts these terms represent.

- “*The burn*” is a broad, colloquial term that refers to intense exercise in which one’s cardiovascular or muscular systems are pushed to their limits. The burn comes from metabolic acidosis, which is produced when you exercise at an intensity that exhausts your muscles’ supply of oxygen. Lactic acid is produced as a byproduct of metabolic acidosis, so a common term you may hear is

lactic acidosis, lactate threshold, or terms associated with lactic. The muscles need oxygen to break down the glucose that produces energy. Without the oxygen, it uses enzymes to break down the glucose, and lactic acid results as a side product. With the muscles overwhelmed and trying to get more oxygen, the bloodstream can become overrun with lactic acid. It is easily overcome with a few minutes of rest.

- *Pain* is a complex phenomenon that occurs for any number of reasons and has influence from biological or physiological, psychological, and sociological factors. The currently prevalent model used to describe an experience of pain is the biopsychosocial model. *Acute* pain may signal—but not always—that something is wrong in the body, such as overuse or an injury. *Chronic* cases of pain are usually complex phenomena where there are no recurring mechanisms of injury and the pain is propagated by various biological, psychological, and/or sociological factors.
- *Soreness* or *delayed onset muscle soreness* (DOMS) is a phenomenon that occurs in response to exercise, stretching, or other activities that the body is not accustomed to. DOMS typically occurs 24 hours after the activity is completed. It is characterized by a dull, indistinct, pain-like sensation in the muscles that is exacerbated by movement. Soreness can be decreased in part by prolonged movement and mobility or some form of light exercise to increase blood flow.

The pain you feel during exercise when your muscles start to “burn” or after exercise is completed does not typically have a negative impact on your body, except in rare situations where it becomes excessive and you continue to push through it. A certain amount of stress on the muscles helps to force anaerobic adaptations in the muscles, a positive goal. This type of “pain” is not an injury risk.

On the other hand, *delayed onset muscle soreness* (DOMS) generally occurs approximately 24 hours after exercise and can last 48-72 hours. In extreme cases, it can last 7-10 days. It typically happens to trained athletes coming back from a layoff. DOMS generally occurs when you (1) try new exercises, (2) increase volume or frequency, or (3) perform excessive amounts of eccentric components of exercises.

Your body is able to progress in both strength and hypertrophy without having to go through the pain of soreness or the burn. Even if you do feel sore, know that as long as you are increasing your strength, gaining muscle mass, or meeting any of your goals you do not have to modify your routine. If you are *not* progressing, *then* it is time to make a strategic change of some sort—even if you are not experiencing any soreness.

Laying out some guidelines for training while sore is a good idea, as people new to training often do not realize the difference between pain and soreness and do not know when they should continue or cease exercising. Here are the guidelines:

- If you are too sore to move, exercise very lightly. You should also be hydrating, self-massaging, foam rolling, or whatever else you can do to alleviate the discomfort. Although studies have indicated that some of these methods do not actually help, the placebo effect can be a strong factor.
- You can work out if you are not too sore, but do not overdo it.
- Otherwise, do not worry about soreness. If you are training frequently enough it should start to go away as your workout capacity increases.
- If you are always sore after workouts, your workout regimen is likely not enough to bring adaptations in the muscles that eliminate soreness (such as 1-2x per week body-part splits). In these

cases, the soreness is probably hindering your workouts. Increase your frequency to 3x per week full-body and the soreness will likely dissipate.

Whether you see it as positive or negative, soreness is not something to worry about, as you can make progress with or without it. Generally, it will be more of an annoyance. If you plan on performing a high-volume workout or a routine with a lot of eccentric movements, you can expect to be sore afterward.

Pain is different. However, chasing pain or a lack of pain is not good. The best concept to understand pain in exercise is the concept of *aggravating exercises*. The reason why pain is less important than aggravating exercises is important. Sometimes if there is pain, an injury may feel and move better after exercise. Sometimes if there is no pain, an injury may get more stiff and worse. Hence, removing aggravating exercises is more important than trying to use pain as a guide. If there is pain, but exercises lessen the pain and improve the injury then keep those. Pain is one factor, but it is not the most important factor in determining whether to use exercises for rehabilitation or not.

For specific advice on whether or not to work through pain, contact your medical professional. A book cannot diagnose you or correct your injuries. You can only learn the factors that affect propensity for injury, learn to integrate these principles for prehabilitation, and learn how to work around injuries in a routine.

FACTORS THAT AFFECT PROPENSITY FOR INJURIES

There are four main factors that contribute to the integrity and health of tissues of the human body:

- *Posture*: The static presentation of the body.
- *Biomechanics*: The movement of the body.
- *Mobility*: The capability of joints and muscles to move within their range of motion.
- *Muscle Length-Tension Relationships*: Interactions that force generation of movement.

These factors all interact with one another. Posture and biomechanics represent the neural control of the central nervous system in its interaction with the environment. *Posture* is the static presentation of the body in space, and *biomechanics* are the dynamic element of movement. We receive feedback from our senses via these pathways in two ways. *Proprioceptive feedback* from static elements, and *kinesthetic feedback* from moving elements. Proprioceptive feedback is being able to sense where the body is in space, and kinesthetic feedback is the ability to sense various movements in space. The brain and its neurological system control the body, and each of these neural elements exerts influence on musculoskeletal structures.

Biomechanics effect constant recalculations of the nervous system as the environment changes. This is why exercises are the best way to teach the body how to move and work correctly. Resistance exercise is also an effective way to train the nervous system and strengthen the muscles. Maintaining proper technique in an exercise significantly reduces the propensity for injury after the body begins to become fatigued.

The rest of the factors all apply broadly to biomechanical movement.

If you have ever been to physical therapy, particularly for back pain, you already know that many of the exercises prescribed are those that work on stability; in effect, re-teaching fundamental movements like those you see babies learning from birth to three years old. As we grow up, our bodies naturally learn movements

that our nervous systems reinforce. However, our culture is one of schools and desk jobs, causing us to “practice” body positions that make us lose our ability to move effectively, thus increasing propensity for injury. Becoming more proficient with rolling, crawling, squatting, and other fundamental movements that our bodies learned when we were younger will drastically decrease propensity for injury due to the body’s inherent carryover effect from simple to more complex movements.

Posture or good body alignment provides a platform to optimize the ability of the musculoskeletal structures to apply and dissipate force when called upon during movement. Studies have shown that bad posture is not always correlated with pain. In other words, there is not always a structural problem in the body where there is pain. MRI studies have shown that 20-50% of healthy, pain-free populations may yet have bulging discs, herniated discs, stenosis (narrowing of a passage that nerves run through), and other degenerative conditions. They have these conditions, even though they do not have pain. Despite these studies, from a health and wellness perspective, good body alignment is a relevant factor to quality movement which may indirectly decrease propensity for injury. This is why we want to take a look at good posture.

Sitting with your shoulders hunched forward (colloquially “caveman posture”) puts your shoulders in an unstable position by changing the length and tension relationships of the muscles. This is prevalent among people who perform desk jobs or work in front of a computer for hours at a time and it causes all kinds of problems. There is the increased propensity for impingement due to internal rotation of the shoulder that leads to decreased space under the acromion. There is increased risk of anterior instability that may lead to subluxation or dislocation due to posterior shoulder weakness. There is increased stress on the AC joint as it moves into the overhead position, which can negatively affect overhead work like handstands. These are only a few examples of increased propensity for injury due to poor posture.

Posture also affects your thoughts and your decision-making. Those who stand taller tend to be more confident and decisive. They are more esteemed by peers for leadership and their voice is given more weight. It is also better for breathing and having a calm mindset in stressful situations.

Your body is trained by the things you do continually. If you teach your body to do negative things, it will learn these things and adjust accordingly—leading to injuries, poor posture, and poor movements. Therefore, prehabilitation and rehabilitative protocol must continually focus on both biomechanics and re-teaching proper posture. Re-teaching your body may require more than simply a few weeks or months of constant diligence to correct problem areas. This is not a chore, but an additional step to improve your overall health and performance.

Mobility is the interaction of the voluntary muscles in their range of motion taking into account how posture and biomechanics affect movement. Kinesthetic feedback is provided to the central nervous system via muscle spindles, which regulate muscle length. Muscle spindles are located in the muscles themselves, while the control system for these muscle spindles is gamma motor neurons. Passive tension is generated when muscles are taken to the edge of their range of motion. Additional muscle length-tension relationships are comprised of kinesthetic feedback to the central nervous system via the golgi tendon organs. These regulate muscle tension and the ability to voluntarily output force. Golgi tendon organs are located at the muscle-tendon junction. Both of these feedback systems are essential to regulate your posture and biomechanics.

Posture, mobility, and the manner in which force is applied all affect how you execute a technique or movement. It is important to focus on these things to build a solid foundation for quality movements. In

terms of managing injuries, an organized approach is best. If you do get injured, use this four-step approach to work your way back to performing a normal routine.

- Goal Setting in Rehabilitation
- Working Around Injuries
- Programming Rehabilitation
- Moving Back into Exercise

GOAL SETTING IN REHABILITATION

Having goals for addressing injuries is important. All physical therapists should have functional goals for their patients. This is required not only for insurance reimbursement, but to warrant the type of treatment that is given. For our purposes, goals give us a timeline. Generally, a medical professional is required to help you set your injury goals because you often do not know how an injury will respond to treatment and how long it will take to heal. You may not even know enough about an injury to know how to rehabilitate it. Do not hesitate to consult a medical professional to come up with goals for your injury—the following guidelines are for informational purposes only.

Goals force you to identify what may or may not be working, and they also keep you motivated. As with your workouts, you should know that something is wrong if you are not progressing from one rehabilitation session to the next on a weekly basis. You may have to make adjustments to your rehabilitation or prehabilitation program to continually make progress in addressing an injury.

In this sense, rehabilitation is a lower-level extension of programming for regular strength and conditioning. It is not that different from regular training. The key is learning the concepts and knowing your options if you are plateauing. Keep in mind that the injured area is sensitive and prone to re-injury. Careful care must be given to an injured body part.

The healing cascade is a normal process of body tissue. It occurs naturally, given enough time. The most prevalent example is with the skin, but all tissues respond similarly when injured—even the brain. The process is predictable and divided into the following three stages:

1. *Inflammation Phase*: Damage stimulates the immune system to clean up injured areas and various growth factors are released to start the proliferation phase.
2. *Proliferation Phase*: New tissues are laid down via rapid cell division.
3. *Maturation Phase*: Tissue is matured and remodeled to handle whatever stresses are necessary in that area.

It is important to realize that the healing cascade is how all body tissues mature to some extent. One of the fundamentals of workouts is that exercise damages muscle cells. Muscles become inflamed and the immune system cleans up the damage. There is proliferation of new muscle cells or satellite cells to repair the damage, and new contractile fibers are laid down. The muscle then matures and remodels to become bigger and stronger.

Another example is that of calluses on the hand. Stress is placed on your tissues when you lift heavy weights or hang onto the pull-up bar to the extent that your hand starts burning and hurting. Your tissues

then become inflamed and begin the process of healing. Your skin cells proliferate and multiply, and they will finally mature and remodel into a callus.

There are problems that can occur with “too much, too soon.” Aggravation of an inflamed area can result in blisters, opening of wounds, scarring, and re-injury. These are so common that you should be able to apply this understanding to your prehabilitation and rehabilitation.

Injuries are best thought of in chronological context. The following timeline can be used to aid you in knowing what to focus on, depending on where you are in the healing process of an injury.

1. *Determine cause of injury.* In bodyweight training, this will most often be a particular movement that aggravates the injured area. For example, elbow tendonitis with pull-ups.
2. *Eliminate aggravating exercises.* Failure to do this is the number one way to make the injury worse. Don't be afraid to remove an exercise from your routine temporarily, even if you are progressing with it. If you don't do this, it will likely hinder your progress in the long run.
3. *Get professional advice from a sports medicine doctor or physical therapist.* This is tied with the first two points. A professional will be able to examine your body for the particular weak link, instability, or other issue that is causing the problem. Knowing exactly what you need to work on will shorten your recovery time.
4. *Follow the advice of your medical professional!* If they do not offer you a plan, you should ask them for one. This is especially true if you are seeing a physical therapist. Ask for exercise(s) you can do at home as well.
5. *Rehabilitate.* If the cause of injury is known, rehabilitation must be integrated with tissue remodeling phases dependent on the diagnosed issue. If your medical professional gives you a plan and exercises to perform, keep these things in mind:
 - Inflammation is a natural occurrence and not inherently bad as long as it is not excessive. Do not try to perform more exercises than your plan recommends. It is important to minimize inflammation on the affected areas.
 - Exercise is important in the healing process; solely resting is the worst thing you can do for an injury. This applies for both acute and chronic conditions.
 - In the tissue inflammatory phase, continuing with exercise is vital in maintenance of range of motion and joint mobility and deterrence of atrophy.
6. *Exercise.* Low weights/intensity, high repetitions, not-to-failure to increase endurance and decrease risk of re-injury, blood flow, and minor amounts of inflammation in the area for healing.
 - The critical point of the tissue proliferation and remodeling phase is keeping a balance between
 - (1) programming in rehabilitation exercises for delicate remodeling of tissues and
 - (2) forcing adaptive changes and organizing remodeling.
 - Exercise must be progressive in correcting issues with posture and/or biomechanics.
 - Exercise must encourage good gross movement patterns.

Tissue healing rates vary depending on which tissue is injured. Skin injuries typically heal in a few weeks, but if you have a cartilage injury or shoulder labrum repair, the proliferation and remodeling phase may last three to

four months since these types of cells heal slowly due to limited blood supply. Functional activities like throwing will not be approved for someone undergoing physical therapy before that amount of time has elapsed.

It is very important to understand what is actually injured and what type of protocol is needed to treat that specific injury. *This is the importance of a professional diagnosis and treatment plan from a medical professional!* Medical diagnosis is essential even if you do not need to go to physical therapy and can self-treat your injury. In the context of prehabilitation and rehabilitation, you should be asking yourself these questions:

- Is what I am doing helping my recovery on a measurable level? (Week to week or month to month.)
- Is what I am doing prolonging my recovery? If so, why? Is it because I am doing too much or not enough? Do I need a professional opinion on whether I am doing too much or not enough?

Use your critical thinking abilities. Do not do something just because it was suggested to you. When in doubt, see a medical professional.

Tissue Type	Quality Trained	Repetitions	Sets	Intensity / % 1 RM
Muscle	Strength	1 – 5	5	80 – 90%
Muscle	Strength & Speed	6 – 15	3 – 5	Fast Pace
Muscle	Endurance & Speed	10 – 25	3 – 5	Fast Pace
Muscle	Strength & Endurance	15 – 30	3 – 5	70 – 80%
Muscle	Endurance	30 – 50	3 – 5	60%
Tendon	Tendon Healing	100 – 200	1 – 3	40 – 60%
Cartilage	Cartilage Healing	Thousands	1	20%

The most important recommendations indicated by the above chart have to do with connective tissue overuse. Though the information may seem counterintuitive, this chart is backed by research and published by the University of Maryland, Baltimore. If you watch the NFL, you often see players with knee ligament or meniscus injuries on the sidelines riding the stationary bike during practice or games. This is because cycling, in particular, is effective for treating these types of injuries for the following reasons:

- *No impact.* Ligaments and cartilage are at a high risk for re-injury with impact because they stabilize the knee during movement. You do not want your last line of defense against injury being tested right after it has been repaired.
- *High-repetition.* This does many things, including increasing blood flow as the joints in your knees are exercised, which allows nutrients to repair injured tissue.
- *Low-intensity.* As the tissue is healing you want to apply low-level mechanical stress to assist in remodeling of the tissues. If you do not do this, your connective tissues may re-align in a cross-hatch pattern rather than linearly, like a rope. Tissues that are not remodeled correctly are at a much higher risk of re-injury.

- *Longer time.* You often spend ten to twenty minutes or more on a cycle. This allows you to accumulate the hundreds, if not thousands, of low-intensity repetitions needed for proper blood flow and mechanical stress without aggravating the area.
- *Not-to-failure.* None of these exercises are performed to the point where you are exhausted, which would stress injured tissues too much and possibly re-injure them.

This is the importance of the high amount of repetitions for tendonitis or other potential connective tissue prehabilitation and rehabilitation. If you suspect your injury is a joint or cartilage issue, have it checked by a medical professional.

There are several effective approaches for overuse prehabilitation or rehabilitation injuries. The general time frame is the same as a mesocycle, four to eight weeks. This tends to be a bit conservative for prehabilitation but is usually spot-on for rehabilitation.

- For most stabilizer muscles, such as the rotator cuff or the muscles around the scapula, employ high repetitions in the 30-50 range, not-to-failure, for 3-5 sets per muscle. Aim to progress slowly in prehabilitation so that the weights do not put you in the failure range for the final repetitions. The goal is to build endurance for the stabilizers, so they have less propensity for injury, especially near failure. This is important because most injuries tend to occur at the beginning of a workout due to improper warm-ups or near the end of a workout when you are fatigued.
- For stabilizer muscles that are used for explosive movements, such as the arm for baseball pitchers, Charles Poliquin's recommendation of 30-100 repetitions per set for endurance training for about 3 weeks followed by phases of 8-12 and 6-8 repetitions at a high tempo are effective. Follow up with specific strength work and explosive training to get re-accustomed to rigorous sports training. This benefits not just baseball pitchers but any sport or discipline with explosive movement like gymnastics, parkour, martial arts, and the like. Your goal is to build endurance to lessen the chance of injury and increase blood flow to heal the area and encourage correct remodeling of tissues.
- There is also a more progressive method that works for most connective tissue overuse injuries or muscles, and it is not limited to stabilizers. Instead of remaining solely in the high-repetition range or jumping straight down to a low-repetition range, gradually decrease the amount of repetitions and increase intensity. For example, work with forty repetitions the first week, thirty repetitions the second week, twenty repetitions the third week, and finally move into the ten-repetition range the fourth week. This represents a gradual increase of intensity to prepare the tissues for the five to twelve range for strength and hypertrophy training, but it also gives the tissues multiple weeks of high-repetition work to increase blood flow, encourage remodeling, and promote overall healing. This can also help the body learn good movement patterns.
- Start with 40% of the volume previously used and add roughly 20% per week until you reach your original workout load. It will take about four weeks to reach full-intensity, giving your injured tissues time to be strengthened. If a relapse occurs, drop the volume 20-40% (depending on how the injury responds) and add 10-15% per week.

These types of exercises respond well to a 2151 tempo, but any similar tempo that varies the time by one second can also work well: 1-3, 0-2, 4-6, 0-2. It is only important to have a controlled concentric component

of 1-3 seconds so as to limit the speed of the repetition in order to ensure that there is enough time to concentrate on proper technique. After a slight pause or rest at the end of the repetition, the 4-6 eccentric phase (muscle lengthening) will re-educate your nervous system to fire correctly. It will also injury-proof the tissue. End with a slight pause or rest at the end of the repetition.

The eccentric tends to be the most useful component in treating any type of pain or injury, especially in prehabilitation and rehabilitation of tendonitis and strains. The eccentric component of exercise causes muscle damage which creates an inflammatory response which is necessary for healing, especially in chronic injuries. The eccentric component also provides sustained, low-intensity stress for remodeling. High repetitions along with muscle contraction help to force blood in and out of the particular areas that need it.

The eccentric component may very well be critical for re-teaching your body to correctly activate its muscles once again, since eccentrics preferentially activate fast-twitch fibers. After a sprained ankle, your body will experience pain, causing your nervous system to sense a threat – which in turn causes your muscles to spasm and tighten to protect the injured area. Eccentrics re-train your nervous system to decrease the threat response and increase motor control again. If you have injured a particular area, you may experience shakiness when trying to control a movement once you begin to perform rehabilitation work. You can correct this with a uniform-speed eccentric movement.

In the physical therapy clinic, eccentrics have also been used to increase flexibility. Patients with frozen shoulders or impaired internal/external rotation have improved 20-30 degrees in range of motion in a single session using not-to-failure slow eccentric motions. The reason for this may be that eccentric movement allows the nervous system to decrease the threat response and re-teach neuromuscular control. The decreased threat response is manifested in a desensitization of gamma motor neurons, which control muscle spindles, which control muscle tension at the edge of the range of motion, usually increasing it. If decreasing your threat response decreases the sensitivity of the muscle spindles, you should be able to move your muscles into a new range of motion.

The theory works well in practice. Slow eccentrics through the range of motion to the edge of the range of motion can be used effectively to gain flexibility or increase range of motion, especially after injury. Aim for a general, systematic approach when addressing a specific area. Here is a typical order of progression for this:

1. Do not aggravate the particular area
2. Regain range of motion and mobility
3. Slowly begin to strengthen the area from all angles
4. Correct any imbalances
5. Move back into regular exercise

Here are specific details for each of these phases. Adjustments can be made depending on how the area heals.

1. Use relaxation and soft tissue techniques on the afflicted area(s). With tendons and connective tissues, it is particularly important to massage the muscles more than the tendons. With overuse injuries, the muscles are typically so tight that even light movements cause them to pull on the attached tissues, thus aggravating the injured area. Loosening the muscles helps significantly.

2. Perform light stretching and mobility work in all directions for the specific tissues that are afflicted. For example, elbow tendonitis will require stretching the flexors and extensors in the wrist, plus mobility work all around. While stretching does not prevent injury, it can be used to help recover from an injury.
3. Train the affected areas with a no impact, high-repetition, low-intensity, not-to-failure method.
4. Strengthen the opposing muscle groups to ensure that there is a balance of strength and hypertrophy at the joints. For example if you have inner elbow tendonitis (golfer's elbow) with your flexors, you will want to perform reverse wrist curls to train your extensors. For anterior shoulder pain, it would be a good idea to strengthen the muscles of your posterior shoulder and scapula.
5. Slowly build up to strength and hypertrophy training from lower progressions.

This approach aims to cover posture, biomechanics, mobility, and muscle length-tension relationships all at once. You may need to add additional elements depending on the injury.

When executing prehabilitation exercises, there are some additional factors you should take into consideration. Adherence to these items will speed your recovery, but keep in mind that there are still physiological limitations of time for healing to fully take place. Here are the factors to consider:

- Prehabilitation and rehabilitation should not aggravate the area where you are working. This should be pretty obvious, but it is important to reiterate.
- Aim to perform any prehabilitative or rehabilitative exercise with a full range of motion and keep control with good form. It is critical to ensure there is no deviation from accurate technique and control when you are recovering from an injury. Rehabilitation is not useful if the wrong structures are being rehabilitated. Additionally, strengthening an area in a limited range of motion is not conducive to regaining full operating capacity of the structure(s) you want to heal.
- During the concentric and eccentric phases of your movements, you are looking for a smooth, well-controlled motion at a uniform speed. Do not worry about acceleration through the movement, like you do in your usual standard strength training. Your biggest focus should be on the slow eccentric phase of the movement, as it is the most important factor in rehabilitation.
- Quit if you are getting close to failure or if the exercises aggravate the injured area. The main goal is to get back to exercising, not to push yourself and potentially get injured again.

Jerky movements lead to instability, especially when repeated often, and may represent a significant factor in the development of injury conditions like tendonitis. They increase impulse forces on the muscles, tendons, and other body tissues, which can also lead to additional damage. It is very important to restore smooth movements through controlled eccentric training. This will allow healing to occur faster and more completely.

Pain is another big inhibitor of muscle function. Whenever your body senses a noxious stimulus, it sends feedback to the motor cortex to inhibit force production. In other words, if you punch someone in the arm and then make them lift weights, their capabilities will be inhibited even if there is not significant damage to their muscles. Your body has natural limiters on the muscle via golgi tendon organs. They limit motor drive if there is pain present.

If you have suffered a catastrophic injury, the amount of time it will take to recover can be estimated by the number of letters in the name of the tissue that is injured. This is fairly easy to remember and gives a

good approximation. For example, a torn ligament will take approximately 8 months (ligament = 8 letters) to rehabilitate properly. Your other tissues are cartilage (9), bone (4), muscle (6), and tendon (6). These figures are used only for catastrophic injuries. Strengthening your tissues to the point where you can exercise again should only take a fraction of this time.

Your overall goal is to reach the point where you can again perform the exercises on your original workout plan. You must progress from low-intensity to high-intensity repetitions of an exercise. Taking a month to get back to full workload is going to be typical for a minimal to moderate injury (around a Grade I strain or two to three months of tendonitis). Decreasing the volume in your routine for a month will allow your tissues to calm down and recover properly. Scale accordingly.

For minimal injuries, it will take a quarter to half of the time that the injury has been present. If the injury has lasted for four months, it will likely take four to eight weeks to reach full health. This is dependent upon individual recovery factors—diet, sleep quality, stress levels, ability to rehabilitate without pain, how your body responds, and more.

This is an extremely conservative protocol, but it is wise to spend a little extra time now so the same injury will be less likely to re-occur. When in doubt, it is always best to err on the conservative side when it comes to volume. You can always increase the volume if needed, but if you attempt to do too much, it is easy to re-injure yourself. Talk to your medical professional and take it slow.

RICE VS. MEAT

There has been a lot of discussion and debate in recent times about a RICE vs. MEAT approach. The following is for informational purposes only; talk to your medical professional before using any of these methods. Here are the definitions of the two acronyms.

Rice

- *Rest*: Rest the injured area.
- *Ice*: Ice the injured area to decrease swelling and inflammation.
- *Compression*: Compress the area (with an ace wrap, sleeve, etc.) to reduce swelling.
- *Elevation*: Elevate the area so the blood in swollen areas can drain back to the heart.

Meat

- *Movement*: Mobilize the area based on your pain tolerance. This assumes that pain is a normal part of an injury and not something to be afraid of. Movement is a critical factor right after an injury has occurred, even over rest.
- *Exercise*: Goes beyond movement to actual exercises aimed at rehabilitating or prehabilitating the injured area. Exclusively no-impact, high-repetition, low-intensity, not-to-failure exercises.
- *Analgesia*: Control the pain with medication instead of ice.
- *Treatment*: Fairly broad term, can encompass all additional methods used in a doctor's office or physical therapy clinic. Could also include alternative medicine, such as cortisone shots, hydrotherapy, contrast baths, ultrasound, acupuncture, dry needling, prolotherapy, plasma rich platelets, extracorporeal shockwave therapy, and more.

The RICE approach is the traditional model for injury treatment. The MEAT approach is an alternative model that is basically physical therapy rehabilitation in disguise. It is important to note that in the MEAT approach, physical therapy is indicated from day one or even day zero. Have a knee replacement? Sprained your ankle? The studies show that earlier you can begin physical therapy, the better. The only cases which are exempted from this are surgeries like ligament or tendon repairs that require anchors or sutures to heal a bit before beginning physical therapy, and any injury that is given a non-weight bearing status.

The RICE approach is rarely used anymore, even for acute injuries like a sprained ankle (assuming there are no catastrophic issues like torn ligaments or broken bones). Rest is obviously not good for a sprained ankle. You want as much early movement as you can get within your pain tolerance level. Slow range of motion movement will actually decrease your pain in many cases. Immobilizing the area leads to much longer recovery times because your muscles begin to stiffen up and motor control quality decreases. The use of ice is hotly debated. In practice, if there is no swelling taking place, ice is unnecessary. If there *is* swelling, compression is actually more effective than ice. Ice is really only useful for dulling pain, and analgesics usually work better for this. If you use compression, elevation is also unnecessary. Based on these factors, the RICE approach would seem to be outdated and inferior to the MEAT approach.

To watch the MEAT approach in action, one need only take notice of those athletes with severely sprained ankles who return to jogging and running the next day when they could not put weight on it for 15-30 minutes after the sprain. For an *inversion sprain* (where the foot rolls inward), work into *dorsiflexion* (moving your toe toward your head) and *everion* (out, to the side, away from your other foot). *Plantar flexion* (moving your toe away from your other foot) and *inversion* (moving the toe toward your other foot) will be painful, since that is how you sprained it. However, if you move slowly and continue to mobilize into dorsiflexion and eversion, the pain will typically begin to subside within ten minutes, and you may even be able to walk on it. If you remain unable to walk, continue to mobilize your ankle with light massage. You should eventually be able to walk without a limp, though your ankle and foot will be sore.

People who use the RICE model often wind up limping or needing assistance to walk, and continue to have difficulty walking for up to a week or more. On the other hand, those who begin mobilization early on, do not use ice, and continue to gently mobilize the injury into the night, tend to wake the next morning feeling fine. They can walk normally and some can even jog or run.

The opinion of a medical professional can not be overstated. If you use the MEAT approach without consulting a medical professional, you do so at your own risk.

WORKING AROUND INJURIES

Working around injuries typically involves removing aggravating exercises in your routine and replacing them with alternative and supplemental exercises until the injury is corrected. The most important thing is that you continue to rehabilitate injuries as you work around them.

Stay in the habit of working out. Intra-limb and opposite limb exercises are tools used for maintaining strength in the joint that has been injured while you continue to normally exercise the unaffected limbs and other areas. This promotes blood circulation which facilitates healing in the areas that have been injured.

In the following example, the athlete has a right shoulder injury. It still hurts to move it, and there are doctor's orders to rest/avoid exercise with this injured body part. Here are some types of exercises you can utilize when working around the injury.

Alternative Exercises

- Aim for alternative exercises. If it hurts to perform dips but it does not hurt to perform pushups, make that substitution. Continue working out without further irritating the injury.
- If a particular exercise irritates an injured area of the body, try moving down a couple of progressions with that exercise. If the lower progression can be performed without irritating the injury, keep exercising using that lower progression.

These are the two main options. It is important to keep working out (safely!) in order to maintain strength and hypertrophy gains and stay in the habit of exercising regularly.

Intra-Limb Exercise

The number one reason your muscles begin to atrophy (the opposite of hypertrophy) is immobility and disuse. Atrophy caused by immobility occurs very rapidly, often as soon as a week or two after restricting a joint due to injury. Anyone who has worn a cast after breaking a bone has seen how this works.

Using the example of the shoulder injury, the first thing you want to examine is whether or not you can use the rest of your arm for exercise. If so, you can counter the effects of atrophy with isolation exercises for your elbows, wrists, and fingers. If a doctor has been seen, it would be a good idea to talk to them to see if arm or forearm exercise is contraindicated during the period of immobility for the shoulder. If exercise is cleared, it is time to get busy.

This is a great time to perform specific isolation exercises for your arms like biceps curls or triceps extensions. Forearm exercises can maintain strength in your arm while your shoulder is immobile. Grip work is often a good choice as well. Exercises that use part of the shoulder muscle for stabilization should be avoided if they irritate the injury.

Using your arm and forearm muscles sends neurological output to the arm as a whole—important, because the radiating effect of neuromuscular impulses will fight atrophy. Activity from the motor neurons is very important to the health of the muscle. In fact, if one were to cut through a motor nerve or sever the spinal cord, related muscles will begin to atrophy within a matter of days. In the case of shoulder injury, you may find it harder to grip the bar during pull-ups or dips due to this effect.

You should aim to do as much light exercise or mobility work as you can without irritating the injured area. It will stimulate blood flow, thus speeding up the healing process and preventing muscle atrophy while maintaining this very important range of motion.

There are almost always ways to continue training injured limbs without making the injury worse. However, if any type of movement irritates an injured joint, this is a major red flag and you should see a medical professional immediately. If your doctor or physical therapist has given you the green light to perform rehabilitation work, but doing so is painful, consult them as well. There are some instances where it is helpful to work through the pain, but this is not always the case. Note that your aim should always be to feel better than when you started exercising, or at least to end feeling good overall.

Opposing Limb Exercise

Referring again to our right shoulder injury model, one of the things that can retain strength in the injured arm is performing strength work with your uninjured arm. Your nervous system is very adaptive; one of its interesting adaptations is the phenomena of *cross education*. When a unilaterally trained skill or strength movement is performed with the opposite limb, strength benefits are transferred, even if that particular limb has not practiced the skill or performed any strength work. Performing unilateral work with your uninjured left arm will benefit your injured right arm.

As an experiment, if you normally throw a ball or dribble a basketball with your left arm, practice throwing or dribbling with your right arm. After a significant unilateral practice, test your left arm. You should be able to dribble better or throw the ball farther without having practiced on that arm at all.

Studies show that the transfer of strength to the opposite limb from cross education is around 5-10%. While this is not much, it is still very beneficial, especially if your injury sets you back more than two weeks. At this point, your body is at risk for muscle atrophy and rapid decrease in neurological strength. Retaining as much strength and stability as possible is critical for speedy recovery.

You do need to take care to ensure that no imbalances develop between uninjured and injured limbs. This can be accomplished by maintaining a low volume and fairly high intensity level, similar to typical strength work. Aim for a few sets of three to six repetitions with the opposite arm if you are going to perform unilateral work.

Unaffected Limbs and Other Areas

After exhausting all options with your injured limb and the opposing limb, you still need to continue working the rest of your body. Some types of full-body exercises like squats and deadlifts may interfere with proper healing. Any exercise that irritates your injury should be avoided until you recover.

This is a great opportunity to work on specific weaknesses that do not interact with your injured right shoulder (to use our model). For example, if you know you have tight hamstrings or a tight back, you can improve these mobility limitations while you are recovering from your injury. This is especially important if flexibility and mobility are lagging behind in certain skills like press handstand, V-sit or manna-type work. For your legs, it may be better to focus on light-intensity exercises to help improve your heart rate and blood flow, as this will accelerate the healing process. Pursue balance and agility work. If your injury is in your lower body you might be able to focus on handstands or rings strength. Work on as many weaknesses as you can without impeding the injury recovery process. Do not allow an injury to steal your motivation; let it be a learning experience and opportunity to work on your weaknesses so you can come back stronger than ever.

FINDING A GOOD MEDICAL PROFESSIONAL

When it comes to finding a good medical professional, you should look for a well-qualified orthopedic doctor, physical therapist, or chiropractor that really listens to your situation and connects with your personality. While there are some massage therapists and personal trainers who know a lot about injuries, this is the exception. Find a professional who works specifically with athletes.

Medical professionals will specialize in different things, especially orthopedic doctors. Some work specifically with the lower body, some with the upper body. There are shoulder specialists, knee specialists, and so

forth. It is less important to find the specialist for your exact injury than to find one that works with athletes and has a good reputation. As the saying goes, *the cream of the crop rises to the top*. These tend to work with college or high school athletic teams and have extensive knowledge of orthopedic injuries and treatment options based on the severity of the injury. Here is a threefold plan for choosing a qualified medical professional:

- Ask your family and friends for a referral. Athletic friends can usually recommend a good doctor, physical therapist, or other medical professionals.
- Contact local schools and universities and ask which physicians and therapists they use for their athletic teams. Chances are they have good knowledge of exercise as they are experienced with athletes.
- If you know any medical professionals, ask them for the name of their doctor or physical therapist. Most health care practitioners know the top professionals in their field, and may even know them personally. There is a much greater chance of success when you see the “doctor of doctors” or “physical therapist of physical therapists.”

Good medical professionals should be able to explain not only why they are performing particular treatments and but also the methodology behind them. If you have any question about a specific treatment, ask them directly. Learn to think critically about your body.

GOOD/BAD EXERCISES AND HOW TO THINK CRITICALLY

Welcome to the world of exercise, where everyone has an opinion on everything. To learn how to discern between “good” and “bad” exercises, follow these steps:

- “Bad” or “worse” is always relative. Ask for specifics.

For example, in terms of shoulder injury risk, one-arm chin-ups are potentially a “worse” exercise compared to weighted pull-ups. This is because in the former, you must control the torqueing movement that is not present with two-handed pull-ups. If the torque is not properly controlled it can lead to rotator cuff dysfunction, tendonitis, or a similar injury.

For something to be “bad” it has to be compared to something else. If someone says a particular exercise is bad, they should be able to suggest a better exercise you can perform instead. Likewise, “good” must be relative to a specific goal. If training for one-arm chin-ups by performing certain exercises is contraindicated, there are other exercises you can use. There is more than one way to reach most goals and exercises fall on a spectrum of good, bad, and everything in between.

- “Bad” or “worse” suggests a mechanism of potential injury.

Using the above example, the mechanism of potential injury is the relative control of the torque at the shoulder. However, you must control torque with any exercise. It is more accurate to say that a one-arm chin-up performed without adequate preparation for effective control is a bad move, rather than labeling the movement itself as bad. For things like squats with a rounded back, the rounded back may increase your risk of injury because the fluid in the disc will start to put posterior pressure on the laminal fibers, which may cause bulging or herniation of the discs. However, rounded back squats or deadlifts can be successfully built up to where there is minimal risk of injury.

Anecdotal evidence and experience counts for some. A coach's story about a patient who randomly hurt their arm doing X is one data point. However, if there are stories of one hundred people hurting their arms doing X the advice tends to be credible. The same is true of research. If research backs up the experience of one hundred people hurting their arms in the same manner, it is safe to label that experience as "bad" for that particular method.

Let's say you have only met a few people who have injured themselves performing kipping pull-ups (not much experience) and there is not a lot of research on the amount of force that kipping pull-ups place on the shoulders (not much data). However, you know there is a mechanism of injury wherein a heavily fatigued body dropping into the bottom of a pull-up position can cause rotator cuff or even labrum injury. This rate of injury appears to be higher in those performing kipping pull-ups than those performing standard pull-ups. You can therefore logically conclude that kipping pull-ups carry a higher rate of injury than standard pull-ups, even though you do not have much experience or research to back it up. In light of this, a good coach would advise beginners or those who are more prone to injury to avoid kipping pull-ups when they are just starting out. This doesn't mean kipping pull-ups are inherently a bad exercise—though they would certainly be bad under certain conditions, such as with beginners (because they have not yet learned proper technique), with those who have previously injured their shoulders, and with athletes who are fatigued.

Exercise is always a cost-benefit assessment. One might perform the most dangerous exercises in the world and never get injured. A cost-benefit assessment looks at exercises as tools. A chain saw is a great tool for cutting, but carries some inherent risk of injury, whereas a power saw or hand saw is less dangerous but also less effective. All exercises bring with them a tension of risk vs. reward, some more than others. If an exercise is considered "dangerous" you may want to avoid it until you are more experienced. Research your exercises so you can be informed of any inherent dangers, and plan accordingly. This is where critical thinking comes in.

Always fact check. There could be a mechanism of injury that sounds logical but is false. Science has known for some time that a gluten allergy is a celiac disease, but has recently discovered that gluten insensitivity appears not to exist. In light of this, one could ask why eating gluten is ever detrimental for people who don't have celiac disease?

(The answer may partially lie in other research such as FODMAPs—fermentable oligo-, di-, monosaccharides and polyols. Those who remove FODMAPs from their diet due to irritable bowel syndrome and other gastrointestinal distress issues may show marked increases in how well they feel and perform. Even though "gluten insensitivity" may be a misnomer, this does not mean certain foods do not cause the symptoms. Wheat and other grain products in particular tend to have a lot of fructans, galactans, and polyols, all of which may irritate the bowels.)

To review:

- "Bad" is relative, especially in setting goals. Seek out alternatives. This can lead to you learning about useful, "good" exercises. If a person criticizes without suggesting a better alternative, they are likely just voicing their personal preference.
- To be truly "bad" an exercise must exhibit potential mechanisms of injury. If no mechanism of injury is indicated, the person advising you against an exercise is most likely spouting "broscience," (which is also colloquially known as "talking out of their butt.") Yes, the word broscience made it into this book.
- It is important to distinguish between hearsay, experience, and research.
- Always fact check. Google, Wikipedia, and WebMD are wonderful tools, but always consult experts in the field as well. This can be more difficult in some areas, such as nutrition.

CHAPTER 15 SUMMARY

HEALTH AND INJURY MANAGEMENT

Propensity for overuse, pain location, tissue quality, posture and positioning, and biomechanics all play critical roles in the development of an injury, either acute or chronic. Even if you have a good idea of what your injury is, have a qualified medical professional check it out.

A general evaluation of the circumstances surrounding any injury brings one to the conclusion that it is always best to see a medical professional for a diagnosis. Once you have a diagnosis, you can directly address the cause of the injury, allowing for faster recovery. If a solution is not readily available, it may be necessary to utilize the “wait and see” approach, though this is not preferred.

A chronological treatment of injury will allow for work on concepts such as protection from atrophy, range of motion, strength, etc. during the inflammatory, proliferation, and remodeling phases.

There are particular loading, repetitions, sets, and volume for rehabilitating specific injuries. Each injury needs to be addressed within the context of its severity and interference with normal bodily function. Most types of injuries can be addressed with light, controlled, eccentric exercises. High repetitions tend to work best.

It is possible to effectively and safely program around injuries using the methods of intra-limb exercises, opposite limb exercises, and exercising in unaffected areas. Intra-limb exercises on unaffected areas help maintain the strength and mobility of uninjured tissues. Opposite-limb exercises assist by the theory of cross education. Exercising in unaffected areas assists with blood flow and overall well-being.

Always perform rehabilitation work for your injuries instead of avoiding them by programming other types of exercises to compensate.

In situations of non-critical injuries, the traditional RICE approach is less effective than the newer MEAT approach. Methods to find a good medical professional and how to approach good/bad exercises by thinking critically were also covered.